CDPM2 UMAT Verification Guide

Tested scope of the distributed source

Document edition 2026-11-09 / Research preview. The tested source is identified by SHA-256 below.

d373a8bee58490c07bccce8a60c16cf3c09b1986c1f484f1d84e6c187192fbd0

This guide contains test methods, results and unmet criteria. Change history is in CDPM2_Change_History.pdf; equations and inputs are in the manual; worked models and results are in the examples guide.

Contents	Page
1. Regression across 312 conditions	2
2. Material-point and tangent tests	3
3. Small finite-element tests	4
4. Restart and scope limitations	5

How to interpret results

Solver completion, history comparison and numerical-criterion qualification are separate outcomes. Completion does not establish constitutive correctness or experimental validity. No fresh native OOFEM campaign was run for this source.

Conditions and evidence

Abaqus/Standard 2026 HF0. Distributed and tested source files are byte-identical. Machine-readable evidence is supplied in evidence/ and VALIDATION.md. Tests requiring private archived point inputs are distinguished from publicly reproducible tests.

1. Regression across 312 conditions

Reference and distributed sources were run with identical environment, materials, viscosity and increment grids. Reference-source SHA-256:

78cc9a681a3526cd1d04e7232ada14ecb18cb116ecd4426b186029ef5536b2de

284 conditions use 240 increments, 26 use 960 and two use 3840. All 36 analyses (18 batches per source) completed with zero MSG errors. Time grids and outputs were compared at all saved times and all eight integration points for 312 conditions.

Quantity	Maximum absolute difference
S	3.81149e-9 MPa
LE	2.97912e-57
SDV14	8.49859e-8
SDV37 / SDV38	1.92469e-10 / 7.59535e-11
SDV39	0
SDV59 / SDV60	1.92423e-10 / 7.58806e-11

Maximum stress difference scaled by max(1 MPa, reference-history maximum) is 3.37009e-11. No appreciable response difference is observed in this scope. No post-hoc acceptance threshold is assigned; comparison status is COMPARED.

This is a UMAT-to-UMAT regression, not an independent OOFEM comparison or proof of global Newton robustness on general structures.

Evidence: evidence/regression312_summary.json; evidence/regression312_comparison.json

2. Material-point and tangent tests

Test	Result
Archived failed Example 2 local attempts	91 / 91 locally converged
Broader archived trials	1094: 724 converged, 370 rejected
Independent spectral test	160 / 160; max relative error 3.1148e-15
Tangent width sweep	700 / 724; 24 unqualified

Tangent criterion

The specified 15 widths from 1e-5 to 1e-12 require errors below 1e-4 at three consecutive widths. The corresponding 11-width grid qualifies 685 points. Counts from different width sets must not be combined into an improvement rate.

At fixed width 1e-9 the maximum error is about 0.832. Perturbations may cross principal-stress sign boundaries; small error is not uniform across points and widths. Trial 947 has protected multiple-root rejection under perturbation.

Reproducibility

Archived replay inputs are private; replay counts are internal archived evidence. The independent spectral test is publicly reproducible. Local convergence is not full-model completion.

```
python tests/spectral_reference.py --compiler gfortran
```

Stubs are only for the standalone test. Never link them into Abaqus or interpret them as validation of Abaqus utilities or parallel safety.

3. Small finite-element tests

The comparator is the preceding characteristic-length-tangent candidate, not the 312-condition reference. S, LE, SDV, U and RF were extracted at all saved frames.

Test	Observed result
C3D8: prescribed traction	40 increments / 40 iterations, no cutback; saved values identical
C3D8: mixed-boundary axial reversal	41 increments, one cutback each, 47 to 46 iterations; max stress difference 1.9001e-5 MPa
Rotated two-C3D8 shear reversal	40 increments / 43 iterations, no cutback; saved values identical
Two-C3D6 wedge shear	40 increments / 44 iterations, no cutback; saved values identical
Serial / requested two-thread / repeat	Saved values identical

These small verification decks use As=15, unlike As=5 in Example 2. Requested two-thread smoke tests are not a race-free proof or MPI validation.

Execution checks

Separate work directories and source/library hashes were used, with successful datacheck before analysis. Execution instructions are in the manual. Document reorganization does not constitute a new solver campaign.

4. Restart and scope limitations

Automatic FIL geometry with 89 state entries and same-source restart completed. Restart used transferred restart/ODB files without the original FIL or external geometry table. Twenty-two common frames were compared.

Quantity	Maximum difference
27 geometry entries	0
S / material SDV	3.58e-13 / 3.56e-13
U	2.63e-18
RF: all 22 frames	5.666e-9
RF: 21 later frames	4.247e-15

Initial-frame RF difference 5.666e-9 exceeds the original absolute 1e-9 criterion. The cause is unresolved; all-frame/all-field equivalence remains not passed. Lack of a persistent later difference does not override the original criterion.

Evidence: evidence/FE_RESTART_RESULTS.json; evidence/RESTART_RF_AUDIT.json

Not established

Production qualification, large-scale/MPI behavior, arbitrary rotations and meshes, general global Newton robustness and experimental validity are not established. Saved examples are not reruns of this distributed source. Independent application-specific checks are required.

Documentation: SIMULIA Abaqus 2026 > User Subroutines > UMAT (initial single-element testing, tangent and state variables), SIMACAESUBRefMap/simasub-c-umat.htm. The official manual does not certify this implementation.